

REPORT 3.1 — The native shell: built, and what the emulator proved (Opus, 2026-09-16)

Job: build docs/native/SPEC_native-shell.md at xhigh, his ruling on 2026-09-16: *"you can run both on x-high and so can Opus for the build. go ahead"*. The job started on his words the same day: *"Let's start the native app build for the tracker now. With the Green flow established at 20ft not 15"* — the green flow is the next job, and nothing here touches putts.

Tree at start: a778194, clean. Four commits, one per Section 11 step:

Step	Commit	What
1	0edbb28	android/tracker/, the web build inside the APK, no INTERNET, Setup screen, WebView
2	168bdfb	K lifted from the bake-off, the native track store, the whole bridge, Kotlin tests
3	704eba7	The web side, Section 6 file by file, browser tests 1-6
4	ce0e04c	The emulator smoke found one real bug: the recorder did not stop at FINISH

42 files, +3,830 / -33. android/bakeoff/* is untouched, and so is every file in Section 7's list: js/gps/gps.js, js/util/geo.js, js/round/track-analysis.js, js/round/round.js, js/analysis/*, sw.js, the export formatVersion, schemaVersion, REVISION, and every IndexedDB code path as it runs on Pages.

Provenance tags as the spec uses them: [measured] read this session from code, a file or a device; [design] a decision; [verify] still owed.

1. The answer, for him

The shell exists, it records, and his rounds come across. One Android app, Golf Tracker, carrying the current screens as its own files, recording the GPS track natively for the length of a round no matter what the phone is doing, and reading his logged rounds back byte for byte out of the exports he already has.

It has not earned a real round yet. The bar is his: 99% coverage and no gap over 20 s, across a round-length carry with the screen locked and a music app in use. That carry has not happened; the emulator proves the plumbing, not the bar. **v27 on Pages stays the instrument** until Fable scores that carry and says PASS in his chat.

One decision is his and it is small: Section 3 below, the two extra points in FT7's track.

2. What is on the phone when this installs

App name	Golf Tracker (the bake-off apps stay GPS Custom / GPS Transistor)
Application id	com.postmaster87.golftracker
Version	versionName v27 , read out of js/data/build.js at build time (D9) [measured]
versionCode	2701 (the web build number x 100, plus a shell counter)
Toolchain	compileSdk 36, minSdk 30, targetSdk 36, AGP 8.13.2, Kotlin 2.3.21, Gradle 8.13, JDK 17 — the bake-off's, pinned [measured]
Permissions	eight, and no INTERNET [measured]
Web files in the APK	30: index.html, manifest.webmanifest, icon.svg, css/ (2), js/ (25, js/dev/ excluded) [measured]
Not in the APK	sw.js, test/, tools/, docs/
APK	8.3 MB debug

aapt dump permissions, the whole list [measured] (Section 9 test 14, **PASS**):

```
ACCESS_FINE_LOCATION      FOREGROUND_SERVICE_LOCATION
ACCESS_COARSE_LOCATION    POST_NOTIFICATIONS
ACCESS_BACKGROUND_LOCATION WAKE_LOCK
FOREGROUND_SERVICE        REQUEST_IGNORE_BATTERY_OPTIMIZATIONS
```

No library merges INTERNET in today (the manifest merger says so), and the manifest carries a tools:node="remove" entry for it so none ever can. That is what makes *"entirely off of webapps and all local"* provable rather than asserted.

3. What Matt must decide

One thing, and it is two GPS points in a round he ditched.

FT7's track (r_1792abc1..., 2026-09-11, the round he shut the app down on) has 4,585 points in the export. Imported into the shell and read back, it returns **4,583**. Nothing was lost on the way in: all 4,585 rows are in the phone's fixes.csv. The reader drops 2 of them because spec Section 4 de-duplicates on the fix's own time, first row wins.

Measured in the source file [measured]: **three byte-identical points share the time 1789165452854** — same latitude, longitude, accuracy, speed and bearing. So the two dropped are copies of one fix, and no measured position is lost. First and last times match. This is exactly the shape REPORT_2.3 Section 11 measured in the field and called an SDK re-emission rather than a new fix.

It is also Section 4 and Section 8 of the spec disagreeing: Section 4 says de-duplicate, Section 8 says every round's read-back count must equal the export's. **Built to Section 4**, because it is a decision with evidence behind it and the alternative is counting one fix three times. The only visible cost: for that one round trackSize reports 4,585 and readTrack returns 4,583.

His call: leave it (recommended), or keep every row and let a repeat count three times.

Everything else in this build is decided. REVISION is not bumped (D10, and the wall in CLAUDE.md Section 3); BUILD.id is not touched; nothing is pushed.

4. Section 9, test by test

Browser suite, /test/ — 531 / 531, all green, n = 3 clean runs at 360x728

519 at v27, plus 12 in one new group, native shell. **Each was proven to fail without its change:** the change was mutated, the suite re-run, the mutation reverted, the suite re-run green.

Mutation	Tests that went RED
timestamp: f.ts becomes Date.now(), and the shim installs with no bridge	3
clearWatch tears the delivery hook down	1
trackstore.js's bridge() returns undefined	4
persistence.js: the shell guard and the shell wording removed	2
schema.js: device.recorder hardcoded 'web'	1
store.js: a stripped payload and no saved path	2

The 12, by spec item:

1. The shim installs only under a bridge; a native fix reaches GpsService carrying **its own time** (Location.time, the fix_ms column), not the arrival time — a receive time there would quietly re-base every gap in every round.
2. clearWatch then watchPosition (the revive path) keeps delivering, and leaves the native watch count where it started (2 up, 1 down).
3. readTrack, writeTrackChunk, trackSize, trackedRoundIds and deleteTrack route to the bridge and keep their shapes; writeTrackChunk filters first and hands over only the valid points; the shell writer stores nothing itself and reports the recorder's rows instead.
4. Persistence reads PERSISTENT in the shell without asking the browser, and says what actually deletes rounds there. The Pages wording is unchanged.
5. newRound().device.recorder is 'web' here and 'native-k' under a bridge, with schemaVersion unmoved.
6. The shell export hands the phone the same payload the download path builds (format, formatVersion, rounds.length, trackPoints), and both export buttons write the file once and say where it went.

Kotlin JVM tests — 10 / 10, n = 1 run

Two files, the bake-off's CoverageTest.kt pattern: TrackFilesTest (6) and RecordingSessionTest (4). Mutation-checked, so they are not vacuous [measured]:

- de-duplicating **last**-wins instead of first-wins: **2 of 10 fail**
- the cap boundary at > instead of >=: **1 of 10 fails**

They cover spec items 7-9: the rounding (7/7/1/2/0 dp), speed absent gives four slots, bearing without speed gives four slots, no accuracy skipped and counted, sample=1 skipped, a duplicate fix time first wins, rows out of order sorted, a cut last row skipped; the import to CSV to read round trip of a 4-, a 5- and a 6-slot point, exactly; and the same-id rule and the 8 h cap on a fake clock.

Emulator smoke — golf-bakeoff AVD, Android 15, 320x640 at 160 dpi

Synthetic GPS at 1 Hz walking north from Veenker's first tee (adb emu geo fix, the bake-off README's recipe). One round, fresh install. **n = 1 each**.

10. HOME 60 s; recents to Clear all; 60 s; reopen. PASS. The process survived the task being removed (the foreground service holds it), the round came back in_progress on the play screen, and the notification never left — 5 samples across the whole closed interval. Fixes across the closed interval (from task_removed to the kill): **72 fixes in 83.6 s, longest interval 1.882 s, zero over 2 s** [measured].

11. kill -9 mid-round; 60 s; reopen. PASS. previous_exit reason=signaled status=9 at 1789571828987, a new process opened the log 1.16 s later, resume_on_process_start, updates_on. **The only gap in the whole round is the 6.044 s spanning the kill**; after it, longest interval 1.714 s, 0 over 2 s [measured].

12. Finish the round. PASS (after the fix in Section 5). recorder_halt, stop, log_close; rows 236 at +6 s and **236 at +21 s**; the notification gone; the service gone; the committed round id dropped. The export written to Download/golf-tracker/ carries **235 points — exactly the 235 data rows in fixes.csv**, 0 cut rows, 0 repeated times, 0 rows without accuracy. The round record carries device.recorder: "native-k", schemaVersion 1, revision 4 — none of them moved.

13. Section 8's acceptance. Section 6 below.

14. No INTERNET. PASS. Section 2 above.

And tools/track-coverage.py reads a shell recorder log as it is, with no change to the tool (Section 4's [verify], now measured):

Source	Span	Fixes	Covered	Gaps > 20 s	Longest gap	Median accuracy	Bar
r_fae282fa...	4.5 min	235	4.5 min (100.0%)	0	6 s	5.0 m	PASS

It read recorder: native-k, the START to STOP window, and named the process death out of previous_exit without being told. The 5.0 m median is the emulator's synthetic value, not a phone measurement.

5. The bug the smoke found, and the one departure it forced

Finishing a round left the recorder running. 319 rows at FINISH, still climbing 12 s later, the committed round id still in prefs and the partial wake lock still held. It would have run to the 8 h cap [measured].

Why: finish and abandon both do the same three things in one tick — set round.status, null ctx.round, then call ctx.stopGps() (js/ui/screen-play.js lines 2510-2530 and 3344-3364) [measured]. So the GPS loop can never see a completed round, and syncNativeRecorder's completed/abandoned branch could not fire from any of the three call sites Section 6 item 2(c) names. The rule was unreachable as written.

DEPARTURE: a fourth call site — stopGps() in js/app.js, which has exactly two callers and both mean "the round is over". D8 is not weakened: a null ctx.round still stops nothing by itself; the **stored** round is consulted, and one still in_progress is left recording. Re-measured on the fixed build: recording stops on FINISH, rows frozen, notification cleared, service gone.

It is not covered by the browser suite — js/app.js is the app shell and the harness never imports it (nothing in test/run.js does) [measured]. Its proof is the emulator, above, and it is the only behaviour in this build whose only proof is the emulator.

6. Section 8: his rounds, moved across

All nine exports in docs/roundDownloads/, imported one at a time through the app's own "Restore from backup" button on the emulator — the real path, the real system file picker, the real confirm sheet — then read back through `GolfNative.readTrack` in one export written through `GolfNative.saveExport`.

On the phone afterwards: **35 rounds, 16 tracks, 59,655 points**, in an 8.0 MB file, written in under 13 s [measured].

14 of the 15 tracks match the source exactly on count, first ts and last ts. The fifteenth is FT7, Section 3 above.

Round	Export	Read back	first/last ts
r_18b4b0bb... (the spec's named round)	8,579	8,579	1789420084966 / 1789427249443, both match
r_49183f52...	11,393	11,393	match
r_b3514a0b...	9,358	9,358	match
r_bfa9018b...	7,858	7,858	match
r_bfa9018b...-fixed	7,858	7,858	match
r_bd6dc819...	5,913	5,913	match
r_1792abc1... (FT7)	4,585	4,583	match — Section 3
r_4770e2b0...	2,327	2,327	match
r_de5532ac...	959	959	match
r_21c09b75...	465	465	match
r_67341063...	66	66	match
r_41fdfed4...	49	49	match
r_98c6639e...	4	4	match
r_79287edb...	3	3	match
r_bbd02109...	5	5	match

The spec's other named values, confirmed against the file [measured]: `golf-tracker-20260914-1807.json` has 6 rounds and 20,090 track points, and every one of those 6 rounds imported at its exact count.

Nine of the fifteen tracks appear in two, three or four of the exports; every copy agrees with every other copy, so there is one right answer per round and the phone has it. Point shapes on the phone after the round trip: 4-, 5- and 6-slot, the same three shapes the exports carry.

An imported track can never be mistaken for a recording: every row carries `provider = import`, `wall_ms` and `elapsed_rt_ms` are empty because this phone never received those fixes, `meta.json` says `recorder: "import"`, and `events.csv` carries one imported line with the point count [measured].

Two of the nine imports needed a harness fix, not an app fix. The picker's search box, once the query had been typed into it, carried the same filename prefix the driving script was matching on, so the script tapped the search box and then tapped RESTORE on nothing. Both files imported first time once the match required a `.json` file row. No app code changed for it.

7. What changed, file by file

New Android project, `android/tracker/` — its own Gradle project beside the frozen bake-off rig.

File	What it is
<code>settings.gradle.kts</code> , <code>build.gradle.kts</code> , <code>gradle.properties</code> , <code>gradlew*</code> , <code>gradle/wrapper/*</code>	The bake-off's pinned toolchain
<code>app/build.gradle.kts</code>	<code>versionName</code> read out of <code>js/data/build.js</code> (D9); the Sync task that copies the web build into the APK's assets
<code>app/src/main/AndroidManifest.xml</code>	Eight permissions, INTERNET removed, portrait, the service
<code>MainActivity.kt</code>	Setup checklist (lifted), the WebView over <code>WebViewAssetLoader</code> , the file chooser, the fix delivery to <code>window.__golfNativeFix</code>
<code>GolfNative.kt</code>	The bridge, Section 5, complete
<code>Recorder.kt</code> , <code>RecorderService.kt</code>	K lifted, plus the three states and the 8 h cap
<code>RecordingSession.kt</code>	The same-id rule and the cap, pure, so both are provable off the phone

File	What it is
SessionLog.kt, Sessions.kt, FixCsv.kt, LocationFix.kt, DeviceState.kt, TrackerApp.kt	Lifted from the bake-off, renamed into the shell's package
TrackFiles.kt	New, pure: fixes.csv to the compact points trackstore.js writes, and back
TrackStore.kt	New: the Android half of trackstore.js
Exporter.kt	The MediaStore copy pattern, lifted
FixBus.kt	The one hop from the recorder to a resumed screen
app/src/test/.../TrackFilesTest.kt, RecordingSessionTest.kt	Section 9 items 7-9
res/...	Icons lifted; the launcher is green, so it can never be mistaken for GPS Custom or GPS Transistor

Web side — every line inert on Pages; the bridge object's presence is the only signal, read at call time.

File	What changed
js/native/shim.js (new)	Native fixes wearing navigator.geolocation's clothes, the way js/dev/sim.js proves the seam. js/gps/gps.js untouched.
js/app.js	Loads the shim; skips the service worker; syncNativeRecorder follows the round; startGps/stopGps skip the wake lock and the pocket lock; stopNativeRecorderIfRoundEnded (Section 5)
js/data/trackstore.js	Every public function routes to the bridge when it is there. Every IndexedDB line untouched.
js/data/persistence.js	PERSISTENT in the shell; a PROTECTED wording that names the right mechanism
js/data/store.js	The export goes out through the bridge and reports the path
js/data/schema.js	round.device.recorder, additive and optional
js/ui/screen-settings.js	EXPORT RECORDER LOGS; the saved path in both toasts; the shell storage wording
js/ui/screen-play.js	The auto-lock control is hidden in the shell
test/run.js, test/index.html	The native shell group
.gitignore	The copied web assets are never committed

8. Departures from the spec, and small decisions

1. **A fourth syncNativeRecorder call site.** Section 5. The spec's three sites cannot see a round end.
2. **The pocket-lock setting is hidden in js/ui/screen-play.js, not js/ui/screen-settings.js.** Section 6 item 7 puts it in Settings; the only such control in the app is the "Auto-lock after" field in the play screen's round menu (line 2419) [measured]. The behaviour is the spec's; the file is the one the control is actually in. Confirmed on the emulator: the round menu in the shell has no auto-lock row.
3. **FT7's two points.** Section 3. Built to Section 4; his call is named.
4. **Section 6 item 2(d) is delivered by item 3.** The spec asks in one place for syncTrackWriter to create no IndexedDB writer and for ctx.trackStats to map GolfNative.stats(), and in another for createTrackWriter in the shell to return exactly that writer. One code path in trackstore.js satisfies both; js/app.js needed no change for it.
5. **"OPEN THE APP ANYWAY" on the Setup screen.** An addition. A checklist that will not let him past — because Samsung reset a battery setting, say — would block logging a round, which the design rule forbids. A missing grant costs track, not shots.
6. **meta.json.** recorder is "native-k" as Section 4 says, so the bake-off's prose string moved to recorder_name. Coverage.kt was not in the lift list, so its two constants (gap_threshold_ms 20000, pass_covered_pct 99) are literals in DeviceState, and the fields stay.
7. **Window insets.** The WebView is inset by the status bar and the gesture bar rather than drawn under them, so the page gets exactly the area Chrome gives a standalone PWA — the area the v26/v27 fold work was measured against (REPORT_2.6: 2340 - 111 - 45, page 728 CSS px). The two bar strips take the page's own theme-color, re-read on page load and on resume.

9. What is NOT in this build

- The green flow. That is the next job and its own spec.
- The course map as app data.
- The watchdog restart from REPORT_2.3 Section 13 — *watch, do not change*.
- Any change to the mark pipeline, the round record's meaning, the export's formatVersion, or REVISION.
- A Pages deploy. Nothing was pushed; BUILD.id did not move.

10. Putting it on his phone

On your phone (Galaxy S26). This is all of it:

1. Settings then Developer options then **USB debugging: On**.
2. Settings then Security and privacy then **Auto Blocker: Off**.
3. Unlock the phone and plug it into the desktop.
4. Tap **Allow** on "Allow USB debugging?".
5. Leave it unlocked on the desk until Claude says it is done.

On the PC, Claude runs every one of these. They are here so they can be checked, not so they can be typed.

SERIAL is the phone's serial from adb devices (RFGL4275NVH), and it must be the phone's, never an emulator's.

```
$env:ANDROID_HOME = "$env:LOCALAPPDATA\Android\Sdk"
cd C:\Temp\gitRepos\golf-tracker\android\tracker
.\gradlew.bat -p . --no-daemon assembleDebug
adb -s SERIAL install -r app\build\outputs\apk\debug\app-debug.apk
adb -s SERIAL shell pm grant com.postmaster87.golftracker android.permission.ACCESS_FINE_LOCATION
adb -s SERIAL shell pm grant com.postmaster87.golftracker android.permission.ACCESS_COARSE_LOCATION
adb -s SERIAL shell pm grant com.postmaster87.golftracker android.permission.ACCESS_BACKGROUND_LOCATION
adb -s SERIAL shell pm grant com.postmaster87.golftracker android.permission.POST_NOTIFICATIONS
adb -s SERIAL shell dumpsys deviceidle whitelist +com.postmaster87.golftracker
adb -s SERIAL shell cmd appops set com.postmaster87.golftracker RUN_ANY_IN_BACKGROUND allow
adb -s SERIAL shell dumpsys package com.postmaster87.golftracker
```

The last line reads every grant back from the system rather than trusting what the grant commands printed. install -r keeps the app's storage, so a reinstall never costs a round.

On your phone again, two things only Samsung can do, and the app cannot read either of them:

6. Open **Golf Tracker**. If the checklist appears, every red row has a **FIX** button next to it; tap them until the list is clear. When it is clear the app goes straight to the round screens on every later launch.
7. Settings then Battery then **Background usage limits**: make sure **Golf Tracker is on neither "Sleeping apps" nor "Deep sleeping apps"**. (An app whose battery is Unrestricted is not offered for "Never auto sleeping apps" — measured on the S26, One UI 8.5, 2026-09-13 — so Unrestricted is the setting, and the two exclude each other.)

Moving your logged rounds across, on your phone, in this order:

8. In **Chrome**, open golf-tracker, then Settings, then **SEND EXPORT...**, and save the file (Drive, or mail it to yourself). **Write down the two numbers in the toast: rounds, and track fixes.**
9. In **Golf Tracker** (the new app), Settings, **Restore from backup**, pick that file, **RESTORE**.
10. Read the toast back. Rounds added and track fixes must equal step 8's numbers. If they do not, stop and say so — nothing has been lost, the export is still the copy that survives everything.

Until the carry is scored, the app is not the instrument. Play rounds on v27 in Chrome, as now. The shell feeds a real round only after Fable scores a round-length locked-screen carry against his bar and says PASS in his chat.

11. Where the evidence lives

- Commits 0edbb28, 168bdfb, 704eba7, ce0e04c on main, not pushed.

- The emulator round and its recorder logs are on the golf-bakeoff AVD (emulator-5556, left running with the app installed and the nine exports in its Download folder, so Fable can re-read any of it). **Nothing was installed on his phone, and nothing on the phone was touched.**
- tools\track-coverage.py scores the shell's logs as they are; the command is python tools\track-coverage.py FOLDER, where FOLDER is what was pulled from Download/golf-tracker/logs.
- Every number in this report was measured this session, on this PC or on that emulator, with its n beside it. Nothing here is recalled.

12. Answered line

2026-09-16 · 3.1 · DONE. The native shell is built to SPEC_native-shell.md: android/tracker/, the web build inside the APK with no INTERNET (8 permissions, measured), versionName v27 read from js/data/build.js (D9), K lifted from the bake-off with the three states and the 8 h cap, the native fix log as the dense track (D2), and the whole GolfNative bridge. Browser suite 531/531, n = 3 clean runs, 12 new tests each proven to fail without its change; Kotlin JVM 10/10, mutation-checked (2 of 10 and 1 of 10). Emulator smoke on golf-bakeoff, n = 1 round: the task closed for 60 s kept 1 Hz (72 fixes in 83.6 s, longest 1.882 s, none over 2 s) with the notification never leaving; kill -9 resumed in 1.16 s with the only gap in the round being the 6.044 s across the kill; FINISH stops the recorder, clears the notification and drops the flag; tools\track-coverage.py reads a shell log unchanged (235 fixes, 100.0%, 0 gaps over 20 s). Section 8 acceptance on all nine real exports through the app's own Restore button: 35 rounds, 16 tracks, 59,655 points read back; 14 of 15 tracks match count, first ts and last ts exactly, including the spec's named r_18b4b0bb... at 8,579 / 1789420084966 / 1789427249443. One bug found and fixed on the emulator: FINISH left the recorder running, because the spec's three syncNativeRecorder call sites cannot see a round end — a fourth site, stopGps(), is the one departure that changes behaviour. Matt's one decision: FT7 reads back 4,583 of 4,585 points, the 2 dropped being byte-identical copies of one fix at 1789165452854 (Section 4's de-duplication); leave it, or count a repeat three times. REVISION not bumped, BUILD.id not touched, nothing pushed. docs/handoff/REPORT_3.1.md.